

Midterm 1

● Graded

Student

TIYA CHOKHANI

Total Points

45 / 50 pts

Question 1

(no title)

20.5 / 25 pts

1.1 (no title)

Resolved 10 / 10 pts

✓ - 0 pts Correct

- 1 pt $g(x) \in [a, b]$

- 1 pt $|g'(x)| < k$ where $x \in (0,1)$

- 1 pt Partial explanation credit

- 1 pt $g(x) \in C[a, b]$

- 2 pts The explanation is incorrect and inconsistent with the prior parts.

- 1 pt The statement is correct but it doesn't indicate what information is in (i) and what information is in (ii) and how they differ.

- 2 pts (iii) explanation is vague.

- 4 pts Missing (iii)

🔄 Regrade Request

Submitted on: Dec 06

wanted to request a small re-evaluation of Question 1.1. I understand that my explanations were marked down for clarity, but I believe the main mathematical content of my answers for parts (i), (ii), and (iii) matches the rubric.

In part (i), I stated that g maps the interval into itself and that g is continuous on the interval.

In part (ii), I stated the standard uniqueness condition, which is that the derivative of g is bounded by a constant less than 1.

In part (iii), although my writing was informal, I did explain that existence must be established before uniqueness, which is the key point the question was asking for.

Since the required conditions were present in my answers, I was hoping you might consider whether the deduction could be reduced slightly for clarity rather than for incorrect or missing content.

I see how you interpreted order. ok.

Reviewed on: Dec 17

1.2 (no title)

In Review 4 / 5 pts

– 0 pts Correct

✓ – 1 pt The approach is correct but the steps are incorrect.

– 2 pts The approach is partially correct, and carried out to its conclusion.

– 3 pts The approach is incorrect, but carried out to its conclusion.

– 3 pts The approach is partially correct and the steps are partially correct.

– 4 pts The approach is unclear or incorrect, and the steps are unclear or incorrect.

– 1 pt missing a conclusion

🔄 Regrade Request

Submitted on: Dec 06

Cancelled Request

 [Edit Request](#)

1.3 (no title)

6.5 / 10 pts

– 0 pts Correct

– 1 pt partial

– 2 pts Partially correct approach carried to its conclusion.

– 3 pts Correct approach with errors in the steps.

✓ – 3.5 pts Partially correct approach with errors in the steps.

– 5 pts Incorrect approach is carried to its conclusion.

– 5 pts Partially correct approach but lacking steps that clearly demonstrate the conclusion.

– 6 pts Partially correct approach but missing any conclusion.

– 6 pts Incorrect approach with incorrect steps.

– 7 pts Incorrect approach with lacking steps.

– 10 pts blank

Question 2

(no title)

24.5 / 25 pts

2.1 (no title)

5 / 5 pts

✓ - 0 pts Correct

- 1.5 pts (ii) does not use monotonicity to make a stronger statement about the roots of f .

- 1 pt (i) defines a root rather than stating the criteria for an existing root given some f .

- 1 pt (ii) does not make a stronger statement about the roots of f .

- 0.5 pts incorrect condition

- 1 pt incorrect statement

2.2 (no title)

Resolved 4.5 / 5 pts

- 0 pts Correct

- 0.5 pts Improper stopping criteria.

- 0.5 pts Lacking clearly defined input.

✓ - 0.5 pts Lacking clearly defined output.

- 0.5 pts Improper loop definition

- 3.5 pts Lacking pseudocode

- 2 pts No newtons step

- 1 pt Incorrect newtons step

🔄 Regrade Request

Submitted on: Dec 06

I wanted to ask if you could reconsider the 0.5 point deduction on Question 2.2 for "lacking clearly defined output." In my pseudocode, I stated P_0 as the input, wrote the Newton update step, and ended with the final P_n value, which was intended to represent the output of the routine. I also included the required assumptions on $f(x)$, as the question asked.

I understand that my formatting may have been informal, but the Newton step, loop structure, stopping condition, and the returned value were all present. Because of this, I wanted to ask whether this small deduction might be reconsidered.

I maintain the original score.

Reviewed on: Dec 17

2.3 (no title)

5 / 5 pts

✓ - 0 pts Correct

- 0.5 pts arithmetic mistake
- 2.5 pts Follow through on Newtons method from 2.b
- 1 pt indicate x_2 greater or less than x^*

2.4 (no title)

10 / 10 pts

✓ - 0 pts Correct

- 0.5 pts Both statements are made although one is not true.
- 1 pt Logic and/or illustration is inverted but reasoning is in the right direction.
- 2 pts The argument mistake is related to the concept of bracketing. The statement is still sound. The illustration is correct.
- 0.5 pts Statement is made for $x_0 \leq x^*$, but the statement is lacking that the first step will be $x_1 \geq x^*$ at which case we can make a statement about $\tilde{x}_0 \geq x^*$. Otherwise logic and illustration is correct.
- 6 pts The statement may be true but is vague, and it's unclear why the statement is true. There is no provided logic and the illustration is unclear.
- 0.5 pts The statement mistake is related to the concept of bracketing. The logic is still sound.
- 3 pts Lacking illustration.
- 1 pt Vague illustration.
- 1 pt Missing a definitive statement.
- 3 pts Statement and logic do not depend on bracketing.
- 2 pts Reasoning is mistaken but the idea is correct.

This test is a closed-book and closed-notes exam. No electric devices or calculators are allowed.

All of your solutions should be easily identifiable and supporting work must be shown. Please clearly label what work you want to be considered for grading. Ambiguous or illegible answers will not be counted as correct.

The duration of the exam is 50 minutes. If you finish early, feel free to submit your exam and leave the room, but not in the last 5 minutes.

Name: Tiya Chokhani

Student ID Number: 305933966

Signature: 

I certify, on my honor, that I have neither given nor received any help, or used any non-permitted resources while completing this evaluation.

1. (25 points) **Fixed Point Damping:** The goal of this problem is to study the damped fixed point method with damping parameter $\lambda \in \mathbb{R}$ in $g(x) = x + \lambda f(x)$, where $f(x^*) = 0$ and $g(x^*) = x^*$.

- (a) (10 points) (i) First state the existence criteria for $g(x)$ to have a fixed point x^* on the interval $[a, b]$. (ii) Then state the criteria for there to exist a unique fixed point on $[a, b]$. (iii) Finally, state the key differences in the order of information required for uniqueness.

i) For $g(x)$ to have a fixed point on $[a, b]$ $g(x) \in C[a, b]$ must be continuous for every $x \in [a, b]$ $g(x) \in [a, b]$ $\forall x \in [a, b]$

ii) For there to be a unique point in $[a, b]$ after conditions stated in i) are met $|g'(x)| < k$ where $0 < k < 1$ for every $x \in [a, b]$

iii) For there to be a unique ^{fixed point} ~~solution~~, i.e. ii) there must first exist a fixed point i.e. i). Thus i) comes before ii). All conditions of i) need to be fulfilled in order ~~to~~ to get to ii)

(b) (5 points) Let $f(x) = x^2 - 3.5x + 2.5$ with $f(1) = 0$ and consider the fixed point method with $g(x) = x + f(x)$ converging to $x^* = 1$. Determine whether the order of convergence is linear. (Hint: You can use a theorem from class to avoid deriving the limit.)

$$g(x) = x + f(x)$$

$$g(1) = 1 = 1 + 0$$

$$\lim_{n \rightarrow \infty} \left| \frac{p_{n+1} - 1}{p_n - 1} \right|$$

$$p_n = g(p_{n-1})$$

$$g(p_{n-1}) = p_{n-1} + p_{n-1}^2 - 3.5p_{n-1} + 2.5$$

$$p_{n+1} = g(p_n)$$

$$\lim_{n \rightarrow \infty} \left| \frac{p_n^2 - 2.5p_n + 1.5}{p_{n-1}^2 - 2.5p_{n-1} + 1.5} \right|$$

Thus the order of convergence is linear as we obtain a linear ~~linear~~ sol

- (c) (10 points) Find the value of λ that achieves the highest order of convergence for $g_\lambda(x) = x + \lambda f(x)$. Justify your answer.
 (Hint: Recall the second in class approach showing the order of convergence for Newton's Method.)

$$g_\lambda(x) = x + \lambda f(x) \\ = x + \lambda(x^2 - 3.5x + 2.5)$$

$$(p - p_0) < \frac{1}{m}$$

$$m = \frac{\max |f''(x)|}{\max |f'(x)|}$$

$$f'(x) = 2x - 3.5 \\ f''(x) = 2 \\ \max |f''(x)| = 2$$

$$p_{n+1} = x_n - \frac{x_n + \lambda(x_n^2 - 3.5x_n + 2.5)}{1 + 2\lambda x_n - 3.5\lambda}$$

we check possible λ values that can maximize the order of convergence for $g_\lambda(x)$

Therefore we get $\lambda = \left(\frac{3.5}{2} - 1\right)^{\frac{1}{2}}$

2. (25 points) **Convexity and Monotonicity and Newton's Method:** The goal of this problem is to study the effects of convexity and monotonicity on Newton's Method. We will focus on a function that is monotonically increasing where $f'(x) > 0$ and convex so that $f''(x) > 0$ for all $x \in [a, b]$.

- (a) (5 points) (i) Clearly state the criteria for the existence of a root of $f(x)$ on the interval $[a, b]$. (ii) Make a stronger assertion for the roots of $f(x)$ by assume that $f(x)$ is monotonically increasing on $[a, b]$. (You do not need to prove your statement.)

i) According to the Intermediate value theorem, if $f(a)f(b) < 0$, aka they have opposing signs there must exist a point $c \in [a, b]$ s.t. $f(c) = 0$, i.e. is a root.

ii) If $f(x)$ is monotonically increasing on $[a, b]$ & $f(a)$ is negative & $f(b)$ is positive there must exist a point c s.t. $f(c)$ is 0 as $f(x)$ is only increasing & it'll be the only ~~point~~ root within $[a, b]$

- (b) (5 points) Use pseudocode to define the Newton's Method routine. Make sure that you clearly define the Newton's step and any requirements on $f(x)$.

$$P_n = P_0$$

$$\text{while } |P_n - P_{n-1}| > \text{TOL}$$

$$P_{n+1} = P_n - \frac{f(P_n)}{f'(P_n)}$$

$$P_n = P_{n+1}$$

~~count~~ count++

- $f(x)$ is continuous & differentiable
- point P_0 isn't too far off from P

- (c) (5 points) Let $f(x) = (x-1)^3 + (x-1)$ use Newton's method to compute x_1 and x_2 starting from $x_0 = 2$. (You do not need to simplify the expression for x_2 , but you should note if $x_2 \leq x^*$ or $x_2 \geq x^*$.)

$$f'(x) = 3(x-1)^2 + 1$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 2 - \frac{f(2)}{4} = \frac{3}{2}$$

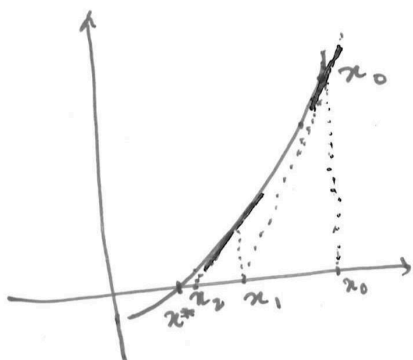
$$f\left(\frac{3}{2}\right) = \frac{1}{8} + \frac{1}{2} = \frac{5}{8}$$

$$f'\left(\frac{3}{2}\right) = \frac{3}{4} + 1 = \frac{7}{4}$$

$$x_2 \geq x^*$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = \frac{3}{2} - \frac{5 \cdot 8}{8 \cdot 14} = \frac{16}{14} = \boxed{\frac{8}{7}}$$

- (d) (10 points) Make a statement about the bracketing of Newton's method for convex and monotonically increasing functions (i.e., $x_0 \geq x_1 \geq x_2 \geq \dots \geq x^*$) that starts with an assumption on the starting side (i.e., $x_0 \geq x^*$ or $x_0 \leq x^*$). Use an illustration to justify your reasoning. (You do not need to prove your statement.)



looking at an example of Newton's method here where we begin with a $x_0 \geq x^*$ then we get the x intercept of the tangent line at $f(x_0)$ & repeat the process.

This results in each iteration of x closer to x^* , i.e.

$$x_0 \geq x_1 \geq x_2 \geq x^*$$

Feel free to use this as scratch.